Year 12 Maths - Proof and mathematical communication

Similar questions - Counter-examples, implication and contradiction

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 For each pair of statements P and Q, state which of $P \Rightarrow Q$, $P \Leftarrow Q$ or $P \Leftrightarrow Q$ is correct, giving a reason.

(a) P: $x^2 = 9$ Q: $x = 3$ **[2]**

(b) P: $\ln x > 0$ Q: $x > 1$ **[2]**

(c) P: n is a multiple of 6 Q: n is a multiple of 3 (where n is an integer) **[2]**

(d) P: $x^2 > 4$ Q: $x > 2$ **[2]**

2

(a) Disprove the statement "If n is a positive integer then $n^2 - n + 41$ is prime" by finding a counter-example. **[2]**

(b) Disprove the statement "The product of two irrational numbers is irrational". **[2]**

(c) Prove by contradiction that the sum of a rational number and an irrational number is irrational. **[4]**

Question 1

(a) $\Leftarrow$: if $x = 3$ then $x^2 = 9$, but $x^2 = 9$ also allows $x = -3$

(b) $\Leftrightarrow$: $\ln x > 0$ exactly when $x > 1$

(c) $\Rightarrow$: every multiple of 6 is a multiple of 3, but $n = 3$ is a multiple of 3 and not of 6

(d) $\Leftarrow$: $x > 2$ gives $x^2 > 4$, but $x = -3$ has $x^2 > 4$ and is not greater than 2

Question 2

(a) $n = 41$: $41^2 - 41 + 41 = 41^2 = 1681$, which is not prime

(b) $\sqrt{2} \times \sqrt{2} = 2$, which is rational (and $\sqrt{2}$ is irrational)

(c) Suppose r is rational, s is irrational and $r + s$ is rational, equal to q. Then $s = q - r$, and the difference of two rationals is rational, contradicting s being irrational. So $r + s$ is irrational